

Operator Manual

Metal Spinning CNC Controller

Documentation

Operator Manual - Metal Spinning Machine

For Machine Operators

Control Panel

[Image not found: img_control_panel.png]

Buttons

Button	Color	Function
START	Green	Begin program
STOP	Red	Stop immediately
PAUSE	Yellow	Pause/Resume
RESET	Blue	Clear errors

Displays

Display	Shows
Status bar	Current state (RUNNING, P
X position	Radial position in mm
Z position	Axial position in mm
Tool	Current tool number (1-4)
RPM	Spindle speed
Feedrate	Speed percentage

Starting a Program

Before Starting [OK]

- ☐ Material loaded correctly
- ☐ Correct tool installed
- ☐ Safety door closed
- ☐ Air pressure OK (green)

[] No error messages

To Start

1. Select recipe (1-5) or load G-code
2. Set feedrate to 100%
3. Press START

During Operation

Speed Control

Use Feedrate Override slider:

- * 100% = Normal speed
- * Lower = Slower, safer
- * Higher = Faster (proven programs only)

Pause/Resume

1. Press PAUSE to stop
2. Machine remembers position
3. Press PAUSE again to continue

Emergency

- * Press red E-STOP button
- * To recover: Release E-Stop -> RESET

Common Errors

Error	Cause	Fix
EMERGENCY STOP!	E-Stop pressed	Release, RESET
Safety Door Open	Door open	Close door, RESET
Air Pressure Low	No air	Check air supply
Axis Move Failed	Motor error	Check obstruction

Hit Limit Switch!	At limit	Jog away, RESET
Motion Timeout	Axis stuck	Check binding

Safety Rules

[!] ALWAYS

- * Keep door closed during operation
- * Wear safety glasses
- * Know where E-Stop is

[X] NEVER

- * Reach into running machine
 - * Bypass door switch
 - * Leave machine unattended (new programs)
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Daily Checks

Before First Run

- ☐ Check lubrication
- ☐ Check air pressure (min 6 bar)
- ☐ Inspect tools
- ☐ Clear debris
- ☐ Test E-Stop

After Last Run

- ☐ Return to home
 - ☐ Clean work area
 - ☐ Report issues
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Status Messages

Message	Meaning
STOPPED - Ready	Ready to start
Homing...	Finding home position
Running	Executing program
PAUSED	Waiting for resume
Tool Change...	Changing tool
Complete!	Finished
ERROR	Problem - see message

Need Help?

Issue	Contact
Programming changes	Maintenance
Mechanical issues	Maintenance
Safety concerns	Supervisor